

Uncovering Covert Racial Stereotypes in Image-Generative AI: a Cross-Platform Multilingual Algorithm Auditing

Background: With the rapid development of image-generative AI, tools like OpenArt and Stable Diffusion are widely used to create realistic and stylized portraits. However, AI is not neutral; it can exacerbate existing biases, both overt and covert. This is critical in image generation, as AI can reinforce societal biases, influencing perceptions of identity, appearance, and culture across racial groups. This paper explores how the language in generation prompts subtly triggers aesthetic stereotypes and reshapes racial traits in generated portraits, thereby reinforcing racial hierarchies, with the aim of providing empirical evidence to help policymakers establish more just and transparent AI regulations.

Research Question: How do linguistic elements in prompt instructions influence the adjustment of facial features and racial traits in image-generative AI, and how do these adjustments indicate racial stereotypes and biases?

Theoretical Framework: Algorithmic bias is a critical concern as algorithms, human-made tools, often reflect and amplify societal and cultural biases. Existing research focuses on overt biases in generative AI, particularly when race, gender, and other attributes are explicitly mentioned in prompts (Sun et al., 2023; Zhou et al., 2024). Most studies on covert biases focus on text-based models, like large language models, which exhibit covert racial biases in processing dialects and linguistic cues. (Hofmann et al., 2024). The study of language attitudes and covert biases in image-generative AI

remains underexplored. Racial characteristics, such as skin tone and face shape, form aesthetic stereotypes (Cross & Cross, 1971), and beauty filters on social media reinforce Eurocentric beauty standards by lightening skin tones and altering facial features (Riccio et al., 2024). However, the relationship between language and AI's visual representation of facial features in portraits has not been fully studied. This study draws on Bonilla-Silva's (2015) concept of colorblind racism and Gumperz's (1982) research on language attitudes to explore whether image-generative AI, in the absence of explicit racial markers, subtly adjusts facial features in portraits based on multilingual prompts, influencing their racial traits. The core objective is to uncover implicit language attitudes in image-generative AI, examine how they shape racial bias, and discuss their implications for global cultural fairness and ethical norms.

Methodology: This study conducts an algorithmic audit using the Fairface image dataset (Karkkainen & Joo, 2021), covering seven racial categories. For each category, a commonly used language was selected, with the following racial categories and corresponding languages: White, Black, East Asian, Southeast Asian, Latinx, Middle Eastern, and Indian, paired with Standard American English(SAE), African American English(AAE), Chinese, Hindi, Spanish, Arabic, and Indonesian. The research uses two popular image-generative AI, Stable Diffusion and OpenArt, ultimately generating 2,940 portraits based on language prompts corresponding to seven races. Face++ facial detection analyzed how AI adjusted facial features, while the Fairface model evaluated the deviations between the racial prediction results of AI-generated portraits and the racial traits of the original images under different language prompts.

Findings: We found that when the prompt language matches the portrait's racial trait, AI subtly alters facial features. For example, Stable Diffusion favors lighter skin tones, while OpenArt lightens skin for Middle Eastern, East Asian, and Southeast Asian portraits, darkens it for Indian portraits. Arabic and Spanish prompts lead to slimmer facial features, while Chinese prompts result in chubby faces. However, both models rarely change the original racial traits. When the prompt language does not align with the racial category, AI adjusts both facial features and racial predictions according to the prompt language. Specifically, Chinese, Hindi, and Indonesian prompts strongly align the generated portrait's racial prediction with the prompt language, while SAE and AAE prompts result in more accurate predictions with minimal change. Arabic and Spanish prompts not only alter the racial prediction to match the prompt language but also misclassify portraits as White, with a higher likelihood of misclassification.

Conclusion: This study provides empirical evidence of colorblind racism in image-generative AI. Even without explicit racial labels, language prompts can activate aesthetic and cultural norms tied to specific races, reinforcing stereotypes. Furthermore, these biases diverge from previous research on social media beauty filters, which predominantly favor Western white aesthetic standards. In contrast, image-generative AI tailors the racial features of generated images based on language-specific cues, with the extent of racial feature variation differing across languages. Users interacting with AI in their native language may be unaware of these hidden discriminatory mechanisms, which inadvertently exacerbate social inequalities. Despite this, the phenomenon of white racial ambiguity in non-English language prompts demonstrates that image-

generative AI still exhibits a noticeable bias toward white aesthetic standards to some degree. We call for greater attention to linguistic and cultural biases in AI to ensure fairness, inclusivity, and ethical integrity in technologies.

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